Tahlia Y. Wu

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EDUCATION

University of Pennsylvania, Philadelphia, PA - GPA: 4.0/4.0 (BSE), 4.0/4.0 (MSE) May 2028
Candidate for BSE in Mechanical Engineering, MSE in Robotics
Relevant Courses: Precision Machine Design and Manufacturing, Design of Mechatronic Systems, Solid Mechanics, Thermodynamics, Dynamics, Linear Algebra and Statics

EXPERIENCE

Mechatronics Engineer
Jan 2025-Present

Aware Lab, University of Pennsylvania, Philadelphia, PA

- Designed and built **FlowBot**, an omnidirectional robotic traverse for automated wind-tunnel wake measurements, independently leading system architecture, CAD, fabrication, and electromechanical integration.
- Implemented real-time **PID control with OptiTrack motion-capture feedback**, achieving **sub-6 mm positioning accuracy** while automating measurements across 650+ spatial locations and 30+ hours of wind-tunnel experiments.
- Designed and assembled a custom **5-ft vertical linear actuator** using 80/20 aluminum extrusion, stepper motors, and precision linear-motion components within a **$200 budget**.
- Modeled **4 custom mounting brackets and mechanical interfaces** in SolidWorks and fabricated functional prototypes using manual machining, laser cutting, and **3D printing**; produced engineering drawings with GD&T and maintained BOM documentation.
- Machined turbine drive shafts to **±0.005 in. tolerances** and conducted **DFM/DFA trade studies**, reducing chassis cost by **53% ($1,500 to $700)** while maintaining design requirements.
- Performed first-principles stress calculations and **structural/modal FEA** to validate a dynamic safety factor > 3.0 and avoidance of wake resonance modes.
- Calibrated **hot-wire anemometers** by calculating and optimizing **sensor overheat ratios** to control operating temperature, maintain thermal stability, and prevent thermal runaway during high-resolution wind-tunnel measurements.
- Integrated hot-wire & sonic anemometers with **LabVIEW using a NI DAQ** and synchronized robotic positioning through authoring a wireless LabVIEW and Raspberry Pi (Python) communication protocol from scratch.

Computational Air Quality Modeling Engineer
May 2025-Jul 2025

Department of Civil, Environmental and Geodetic Engineering, Ohio State University, Columbus, OH

- Developed a scalable roadside pollutant dispersion model using EPA MOVES/R-LINE, extending predictions across **2,500 m road corridors without rerunning simulations**.
- Processed **2,000+ real-world road** links using k-means clustering and developed a superposition algorithm to model emissions across multiple road segments and wind conditions.
- Presented research findings to **100+ attendees** at the OSU Summer Symposium and co-authored a research framework submitted to ACS ES&T Air.

Computer Vision R&D Engineer
Jun 2023-Oct 2023

Davis Lab of Computer Vision, UC Santa Cruz, Santa Cruz, CA

- Led a **3-engineer team** developing a 3D point-cloud mapping pipeline for spatial localization and tracking of **468 facial landmarks** for use in automating preprocessing of training data for reinforcement learning.
- Developed Python/C data-processing and evaluation pipelines using **kd-tree nearest-neighbor interpolation** and geometry-based filtering to improve spatial alignment and measurement reliability.
- Validated localization accuracy across **129 3D scans**, achieving **1.03 ± 0.08% coordinate error** and benchmarking performance across complex spatial geometries

PUBLICATIONS & PRESENTATIONS

Speaker *"Robotic Flow Measurements of Unsteady Wind-Turbine Wakes"* (Accepted in APS Division of Fluid Dynamics, 2026)
2nd-Author *"A Scalable Framework for Spatiotemporal Modeling of Traffic-Related Emissions"* (Submitted to ACS ES&T Air, 2026)

SKILLS

CAD & Manufacturing Processes: DFx (DFM/DFA), Design validation & reliability testing, Parametric CAD - SolidWorks (CSWA Certified), Mastercam, CNC Machining, Manual Mills & Lathes, 3D printing, Laser cutting, Rapid prototyping, Tolerance stack-up, Soldering, Engineering Drawings, Bills of Materials (BOMs), GD&T, FEA (structural, modal)
Robotics & Controls: PID Control, Kinematics, OptiTrack Motion Capture, Raspberry Pi, Arduino, PWM, Electromechanical integration, Oscilloscopes
Software: Python, C, C++, Java, R, HTML/CSS, LabVIEW, Github, PyTorch, Computer Vision, MATLAB, Microsoft Office Suite, SQL